Advanced Econometrics – Lab 08 – Mock test

Exercise 1

"The data in JTRAIN98 are on male workers that can be used to evaluate a job training program, where the variable we would like to explain, $y = earn98$ is labor market earnings in 1998, the year following the job training program (which took place in 1997). The earnings variable is measured in thousands of dollars. The variable *train* is the binary participation indicator. Participation in the job training program was partly based on past labor market outcomes and is partly voluntary. Therefore, random assingment is unlikely to be a good assumption. As control variables, we use earnings in 1996 (the year before the program), years of schooling (*educ*), age, and marital status (*married*). Like the training indicator, marital status is coded as a binary variable, where $married = 1$ means the man is married."[1]

a) Why shouldn't we use simple regression model for the problem? Explain.

b) Is this a corner solution or censoring variable case? Explain.

c) What is the interval of values of the dependent variable? How many observations did take values on the edge of the interval?

d) Which variables are individually significant? Use 5% significance level.

e) Interpret marginal effects in the column `E(y|x)`.

f) Interpret marginal effects in the column `E(y|x,y>0)`.

g) Interpret marginal effects in the column `Pr(y>0|x)`.

```
Call:
censReg(formula = earn98 ~ train + earn96 + educ + married, left = 0,
    right = Inf, data = jtrain98)

Observations:
         Total  Left-censored     Uncensored Right-censored
          1130            194            936              0

Coefficients:
            Estimate Std. error t value  Pr(> t)
(Intercept) -6.49981    1.02173  -6.362 2.00e-10 ***
train        3.75489    0.53702   6.992 2.71e-12 ***
earn96       0.49204    0.02266  21.712  < 2e-16 ***
educ         0.70048    0.07480   9.364  < 2e-16 ***
married      1.01752    0.49037   2.075    0.038 *
logSigma     1.96830    0.02390  82.346  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Newton-Raphson maximisation, 7 iterations
Return code 2: successive function values within tolerance limit
Log-likelihood: -3346.594 on 6 Df
```

[1] Source: J. M. Wooldridge, *Introductory Econometrics, A Modern Approach, 7ed., p. 101-102.*

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```
> x = rbind(1, 0, mean(jtrain98$earn96),
+           mean(jtrain98$educ), 0)
> me = tobit_marginal_effects(tobit, x, dummies_indices=c(2,5))

 Marginal effects of the tobit model
 -----------------------------------
               y*    E(y|x) E(y|x,y>0)  Pr(y>0|x)    at X=
train!   3.7548937 0.8735787  0.6631566 0.03183509  0.00000
earn96   0.4920415 0.4144034  0.3103544 0.01657222 12.19998
educ     0.7004805 0.5899532  0.4418269 0.02359255 10.96549
married! 1.0175230 0.8569703  0.6418010 0.03427072  0.00000
(!) indicates marginal effect was calculated for discrete change
of dummy variable from 0 to 1
```

R code to repeat the analysis:

```
install.packages("wooldridge")
library("wooldridge")

data('jtrain98')
str(jtrain98)
library( "censReg" )

# Tobit model
tobit <- censReg(earn98~train+earn96+educ+married,
                left=0, right=Inf, data=jtrain98)
summary(tobit)

# marginal effects after censReg function
source("tobit_marginal_effects.R")
x = rbind(1, 0, mean(jtrain98$earn96), mean(jtrain98$educ), 0)
me = tobit_marginal_effects(tobit, x, dummies_indices=c(2,5))
```

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